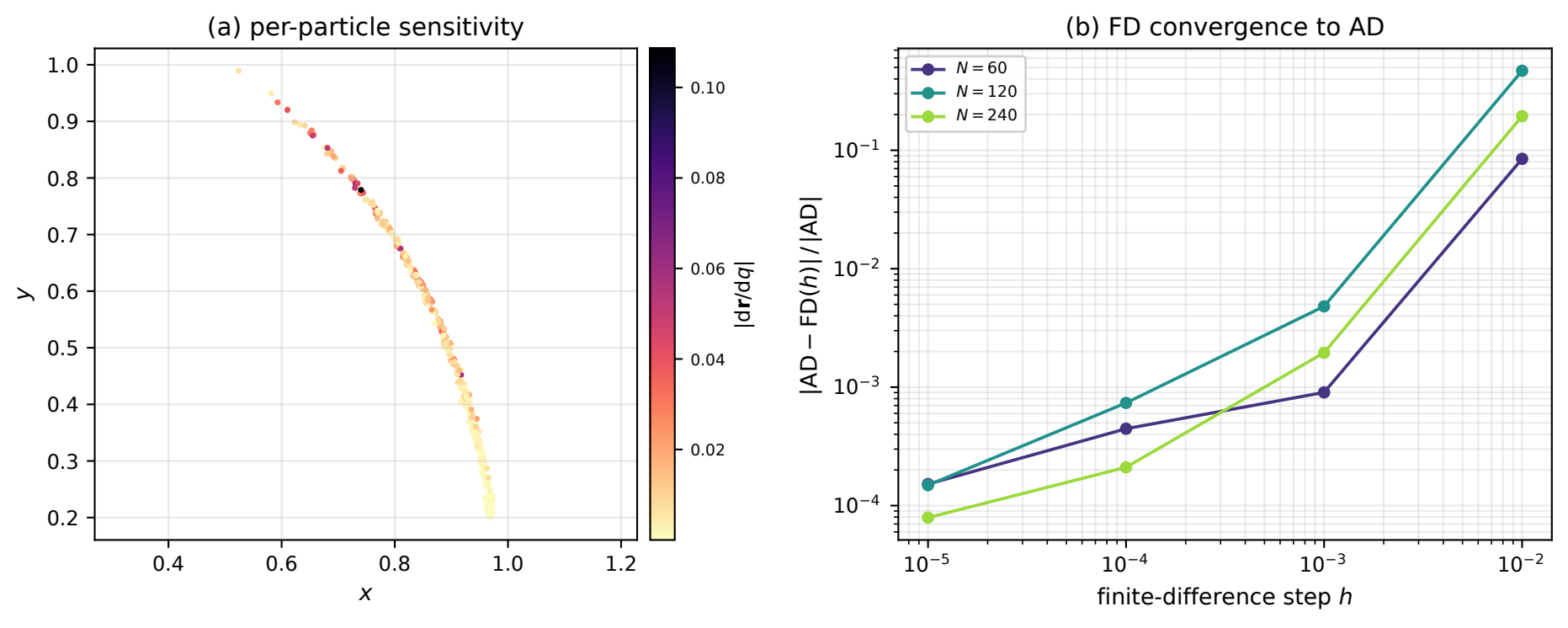
(a) per-particle sensitivity
x
y
|dr/dq|
(b) FD convergence to AD
N = 60
N = 120
N = 240
|AD − FD(h)| / |AD|
finite-difference step h